

Noether's theorem: Conservation laws and symmetries

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1 Classical mechanics

- For N -particle system with C holonomic constraints $f(r_j, t) = 0$, describe configuration with $n = 3N - C$ **generalised coordinates** $q(t) = (q_1(t), \dots, q_n(t))$
- Associated **kinetic** and **potential energy** of configuration, $T(q, \dot{q}, t)$ and $V(q, t)$
- Define **Lagrangian** and **action**

$$L = T - V, \quad S = \int L dt.$$

- Principle of Least Action:** Motion of a system governed by **minimising action** S , and so satisfies the **Euler-Lagrange (E-L) equation**:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) = \frac{\partial L}{\partial q_i}$$

- Noether's theorem:** If L is **invariant** under a **continuous symmetry** (e.g. spatial/temporal translations, but not reflections), there is a **first integral** (i.e. a conserved quantity), which utilises the E-L equation.
- For example, if L is independent of q_i , then we have

$$\frac{\partial L}{\partial q_i} = 0 \Rightarrow p_i = \frac{\partial L}{\partial \dot{q}_i} \text{ constant}$$

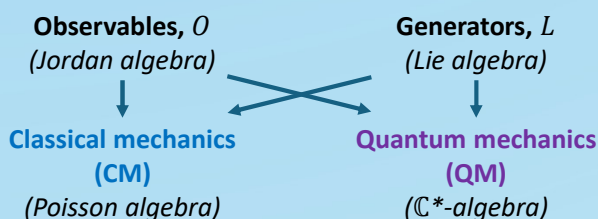
- p_i are the **conjugate momenta** associated with q_i
- We can observe this **"duality"**:

Invariance	Conservation
Spatial translation	Linear momentum
Rotation	Angular momentum
Temporal translation	Total energy

- There are other **conserved quantities** derived from **symmetries** (specifically, **gauge transformations**):
 - Electric charge
 - Colour charge
 - Difference between baryon and lepton number

2 An abstraction

- We can view mechanics via algebraic structures:



3 Symmetry in algebraic structure

- For **classical mechanics**, use **Poisson algebra** A , equipped with **Poisson bracket** $\{*,*\}$.
- For **quantum mechanics**, use **$\mathbb{C}^*$ -algebra** A , equipped with **commutator** $[*,*]$.
- $a \in A$ generates **one-parameter family of transformations** F_t^a if there exists **unique solution** to **Hamilton's/Heisenberg's equation** for each $b \in A$:

$$\frac{d}{dt} F_t^a(b) = \{a, F_t^a(b)\}, \quad F_0^a(b) = b$$

$$\frac{d}{dt} F_t^a(b) = [ia, F_t^a(b)], \quad F_0^a(b) = b$$

- a is a **symmetry** of b if $F_t^a(b) = b \quad \forall t$.
- Euler-Lagrange $\rightarrow$ Hamilton's/Heisenberg's equation
- In Poisson algebra, each element naturally **act as generator and observable**.
- In $\mathbb{C}^*$ -algebra, **generator-observable identification**

$$\psi: O \rightarrow L, \quad a \rightarrow ia$$

but O and L are **distinct** (in fact, set of self- and skew-adjoint elements resp.)

- Note other views of QM (e.g. using Hilbert spaces over $\mathbb{R}^n$ or $\mathbb{H}^n$) do **not** have identification ψ :
 - Over $\mathbb{R}^n$, $\frac{n(n+1)}{2} = |O| > |L| = \frac{n(n-1)}{2}$
 - Over $\mathbb{H}^n$, $2n - n^2 = |O| < |L| = 2n^2 + n$

4 Noether's theorem

- Noether's theorem hinges on
 - anti-symmetry of **Poisson bracket** $\{*,*\}$ in CM or **commutator** $[*,*]$ in QM
 - identifying generators with observables

- Noether's theorem:**

$$a \text{ symmetry of } b \Leftrightarrow b \text{ symmetry of } a$$

$$\{a, F_t^a(b)\} \Leftrightarrow \{a, b\} \Leftrightarrow \{b, a\} \Leftrightarrow \{b, F_t^b(a)\}$$

$$\{ia, F_t^a(b)\} \Leftrightarrow \{a, b\} \Leftrightarrow \{b, a\} \Leftrightarrow \{ib, F_t^b(a)\}$$

- "Self-conservation principle":** each observable generates symmetries of itself.

$$\{a, a\} = 0$$

References

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